
Diagnosing MAPPO with Communication in a Pursuit-Evasion Game

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Abstract

Multi-agent reinforcement learning provides a useful framework for studying coordination in complex environments. In this work, we investigate two key challenges in the pursuit–evasion setting: credit assignment and the role of communication. To address credit assignment, we redesign the reward structure with individual capture rewards, distance-based shaping, and obstacle collision penalties, providing more localized feedback and improving training stability. We further incorporate a communication channel and evaluate its impact through bandwidth variation, cutoff intervention, and per-speak cost experiments. While communication improves performance over the no-communication baseline, its benefit quickly saturates, and agents perform comparably when communication is disabled. These results suggest that in fully observable environments, coordination can largely emerge without explicit communication, as relevant information is already available through local observations.

1 Introduction

Early work Littman [1994] introduced the framework of multi-agent reinforcement learning, highlighting the complexity of interactions among multiple agents. A comprehensive survey Buşoniu et al. [2008] further identified key challenges such as non-stationarity, partial observability, and scalability. Multi-Agent Proximal Policy Optimization (MAPPO) Yu et al. [2021] is designed to address these problems and has been widely applied in a variety of multi-agent scenarios such as StarCraft, Google Research Football, Hanabi, and coordinating heterogeneous robotic teams.

Beyond these applications, learning effective coordination in multi-agent systems remains challenging. In particular, assigning proper credit to agents when they play different roles is non-trivial, since the overall performance depends on collective behavior. Designing proper coordination mechanisms is also important. Communication is often considered a key approach to address these challenges, but it is still unclear when and how communication actually improves performance.

To study these problems, we adopt a simplified pursuit-evasion game as our experimental environment, in which multiple agents cooperate to capture a single target agent. We use MAPPO as our baseline algorithm. To enable communication, we extend the action space by introducing a discrete communication channel. At each timestep, each agent generates communication messages and shares them with others. These messages are incorporated as additional inputs to the neural network policies, allowing agents to learn how to both send and utilize information.

In addition, we redesign the reward structure to address the credit assignment problem, enhancing cooperation by providing different credit signals for agents playing different roles. We also introduce a per-speak cost on the communication action and sweep its magnitude to test whether the policy uses the channel out of necessity. Finally, we conduct ablation studies using communication cutoff and retraining strategies to evaluate whether communication truly improves cooperation efficiency.

Overall, this work aims to find a proper way to assign the credits to different agents and systematically investigate whether explicit communication can improve coordination efficiency in cooperative multi-agent systems, and under what conditions it is most effective.

Concretely, we ask: (Q1) does communication improve return over a no-communication baseline, and how does the gain scale with bandwidth; (Q2) does the trained policy depend on message *content*, or only on message presence; (Q3) what task-relevant information do messages encode; (Q4) does role-like behavioral organization require the channel; and (Q5) can communication activity be reduced without performance loss? Across all five diagnostics, the channel encodes some prey-spatial information but is used opportunistically rather than necessarily; coordination is largely supported by the movement policy and positional observations.

2 Related Works

In recent years, multi-agent systems have made significant progress in many scenarios. However, there are still some important challenges in collaborative multi-agent systems, such as credit assignment and the effectiveness of communication mechanisms.

For credit assignment, existing approaches primarily operate at the learner level. For example, COMA Foerster et al. [2018] introduces a counterfactual baseline to isolate each agent’s marginal contribution, while VDN and QMIX Rashid et al. [2018] decompose the joint value function into per-agent components to achieve implicit credit assignment. In contrast, we adopt a more direct approach at the reward level: by correcting issues in the `simple_tag` reward design and introducing a hybrid individual/team reward scheme, we provide more localized and role-relevant feedback without modifying the MAPPO architecture.

For communication mechanism, previous work such as CommNet Sukhbaatar et al. [2016] and DIAL Foerster et al. [2016] introduced differentiable communication mechanisms. Subsequent methods, including TarMAC Das et al. [2019] and IC3Net Singh et al. [2019], enhanced expressiveness through attention and gating mechanisms. Based on the results of this series of work, more recent studies focus on communication efficiency—specifically, *when* and *how much* to communicate. For instance, NDQ Wang et al. [2020] leverages an information bottleneck objective to learn minimal yet sufficient messages, while Karten et al. [2023] apply sparsity penalties to reduce redundant communication.

Instead of proposing a new communication mechanism, we adopt a simpler design: a fixed-dimensional discrete broadcast channel with an explicit null token. On top of this, we introduce a *per-speak cost* sweep to analyze the functional role of communication in decision-making.

Furthermore, recent studies have raised doubts about whether the performance improvements achieved through communication truly reflect the causal use of information content. Lowe et al. [2019] argued that merely improving the return values is insufficient as evidence and proposed using counterfactual intervention to evaluate the effect of communication. Subsequent studies further combined auxiliary loss and linear probing techniques to analyze the learned representations. Following this approach, we combined counterfactual intervention with representation analysis to systematically evaluate the communication situation in MAPPO. Our research results indicate that although the communication channel encodes useful information, its impact on behavior is limited in an environment that is almost completely observable.

3 Background & Preliminaries

3.1 Multi-Agent Reinforcement Learning

Multi-agent reinforcement learning (MARL) studies an environment where multiple agents interact with each other and learn simultaneously. Each agent aims to maximize its own expected return, and the dynamic changes of the environment depend on the joint actions of all agents. Therefore, from the perspective of a single agent, the environment becomes unstable because the behavior of other agents will constantly change during the training process. This makes it more difficult to learn stable and coordinated strategies compared to single-agent settings.

3.2 Multi-Agent Proximal Policy Optimization (MAPPO)

Multi-Agent Proximal Policy Optimization (MAPPO) extends Proximal Policy Optimization (PPO) to multi-agent scenarios, adopting a centralized training and decentralized execution mode. During the training process, a centralized value function can obtain global information from all agents, which helps to stabilize the learning process in non-stationary conditions. In the execution stage, each agent acts independently based on its own local observation results. Due to its simple structure and excellent empirical performance, MAPPO has become a widely used benchmark model for collaborative multi-agent tasks.

3.3 Communication in Multi-Agent Systems

In multi-agent reinforcement learning, communication mechanisms are often introduced to enhance the collaboration among different agents. In implicit communication, agents infer relevant information by observing the behaviors of other agents; while in explicit communication, agents exchange information through dedicated communication channels. In this study, we focused on the explicit communication method, where each agent generates information at each time step and shares it with others. This information is incorporated into the additional input of each agent’s strategy, enabling the agents to learn how to send and utilize information during the training process.

4 Method

4.1 Problem Setup

We model the task as a fully observable Markov game $\langle N, \mathcal{S}, \{\mathcal{O}_i\}_{i=1}^N, \{\mathcal{A}_i\}_{i=1}^N, P, \{R_i\}_{i=1}^N, \gamma \rangle$. The local observation $o_i^t \in \mathcal{O}_i$ includes each agent’s own state and the relative positions and velocities of other agents and landmarks. The action space is $\mathcal{A}_i = \mathcal{A}_i^{\text{move}} \times \mathcal{A}_i^{\text{comm}}$, implemented as a `MultiDiscrete` action space with movement branch $|\mathcal{A}^{\text{move}}| = 5$ and communication branch $|\mathcal{A}^{\text{comm}}| = K$. Rewards R_i are defined by agent role in Section 4.1.1, and the discount factor is set to $\gamma = 0.99$.

We use the `simple_tag` predator-prey environment from MPE Lowe et al. [2017] with three predators (*adversaries*, agents 0–2) and one prey (*good agent*, agent 3) in a continuous 2D arena, along with two static collidable landmarks. Physical parameters are listed in Table 1: the prey is faster ($v_{\max} = 1.3$, $a = 4.0$) and smaller ($r = 0.05$) than each predator ($v_{\max} = 1.0$, $a = 3.0$, $r = 0.075$), so capture requires multi-predator coordination.

At the start of each episode, all agent positions are sampled uniformly from $[-1, 1]^2$ with zero initial velocity and a zero communication state. Each agent observes its own absolute position and velocity, the relative positions of all other agents and landmarks, the absolute velocity of the prey, and (when communication is enabled) the communication states of the other predators (§4.2).

Agent	Role	Size	Acceleration	Max Speed
Agent 0	Predator	0.075	3.0	1.0
Agent 1	Predator	0.075	3.0	1.0
Agent 2	Predator	0.075	3.0	1.0
Agent 3	Prey	0.05	4.0	1.3

Table 1: Physical parameters of agents in the `simple_tag` environment.

The prey is subject to a boundary penalty $b(x) = \max(0, 10(|x| - 0.9))$, clipped at 10 on each axis x , discouraging escape beyond $|x| > 1$.

4.1.1 Credit Assignment in Multi-Agent Reward Design

The vanilla `simple_tag` reward function in the MPE codebase contains two implementation bugs that conflate per-predator contributions with team-level outcomes, breaking credit assignment for individual MAPPO learners.

Bug 1: Distance shaping summed across predators. The vanilla `adversary_reward`, although called once per agent to compute that agent’s reward, loops over *all* adversaries when computing the distance-shaping term:

$$r_{i,t}^{\text{shape,vanilla}} = -0.1 \sum_{j \in \text{predators}} \min_{k \in \text{prey}} \|p_t^j - p_t^k\|_2,$$

yielding a value that is identical across all three predators. Each predator therefore receives a shaping signal proportional to the *team*’s aggregate distance to the prey rather than its own.

Bug 2: Capture reward broadcast to all predators. The vanilla capture-reward block iterates over all (prey, predator) pairs, so a single capture by predator j^* awards +50 to every predator i , not only j^* :

$$r_{i,t}^{\text{capture,vanilla}} = 50 \cdot \mathbb{I}(\exists k \in \text{prey}, j \in \text{predators} : \text{collision}(k, j)).$$

This converts a per-agent capture signal into a team signal and removes any incentive for the non-capturing predators to position themselves for the capture.

Together, these bugs make the trained policies effectively share a fully-symmetric reward, eliminating the per-agent learning signal that MAPPO’s separated advantage estimates are supposed to consume.

Fixes. We patch both terms to be per-agent. Distance shaping uses only the calling agent’s own distance, $r_{i,t}^{\text{shape}} = -0.1 \min_k \|p_t^i - p_t^k\|_2$. The capture reward uses a mixed self-credit / team-credit scheme:

$$r_{i,t}^{\text{capture}} = \begin{cases} +50 & \text{if predator } i \text{ collides with the prey,} \\ +10 & \text{else if any other predator collides with the prey,} \\ 0 & \text{otherwise.} \end{cases}$$

The +10 teammate-credit term keeps a (small) cooperative incentive for predators that contribute to capture without making the final hit, while $+50 \gg +10$ preserves a strong individual signal for the actual capturer. We also add a -1.0 penalty per predator-landmark collision, absent in the vanilla code. Prey capture and boundary penalties are unchanged, while we additionally penalize prey-landmark collisions by -1.0 .

4.2 MAPPO with Communication

Each predator has its own MAPPO actor [Yu et al., 2021] with separate categorical branches for movement and communication: a movement head over the original 5 discrete directions (no-op, $\pm x$, $\pm y$) and a communication head over a vocabulary of $K = \text{comm_dim}$ tokens (§4.2.1). The two heads are sampled independently, so the joint action $u_t^i = (a_t^i, c_t^i)$ has factorized log-probability $\log \pi(u_t^i | o_t^i) = \log \pi_a(a_t^i | o_t^i) + \log \pi_c(c_t^i | o_t^i)$; PPO clipping and the entropy bonus are computed on this joint quantity. Critic and reward inputs follow standard CTDE-MAPPO. The prey is silent (`comm_target=adversaries`); only predators emit and receive messages.

A predator’s emitted token at step t is converted to a one-hot vector and written to its environment-side communication state $c_t^i \in \mathbb{R}^K$. Communication is *undirected broadcast*: the next observation of predator j contains $\{c_t^k\}_{k \neq j}$ concatenated with its physical observation, so messages affect listeners with one environment step of latency. The full reward and per-speak cost terms applied during training are described in §4.1.1 and §4.2.3.

4.2.1 Communication Effect (Bandwidth Sweep)

We sweep the communication dimension $K = \text{comm_dim} \in \{0, 2, 4, 8, 16, 32, 64\}$, where each predator’s per-step communication action is a discrete token $c_t^i \in \{0, 1, \dots, K-1\}$ (with $K=0$ corresponding to no communication channel, recovering the no-comm MAPPO baseline). In the bandwidth sweep without speak cost, all K communication tokens are ordinary one-hot messages. A separate null token is introduced only in the per-speak-cost experiments. Three independent training seeds are run per K , all using the shared MAPPO hyperparameters specified in §4.2. Reported metrics including mean predator return, mean first-capture step, and capture rate, evaluated against both random and heuristic prey opponents are shared across all subsequent subsections (§4.2.2–§4.2.4).

4.2.2 Communication Necessity (Cutoff Intervention)

To test whether the trained policy’s task performance depends on message contents, we apply five test-time interventions to the predators’ communication slots, following Lowe et al. [2019]. The trained policy is frozen and predator movement actions a_t^i are unchanged; only the communication slot c_t^i is overwritten between policy output and environment step.

The interventions are: *vanilla* (no intervention; reference for ΔR); *zero* (replace with the all-zero vector, removing all message information); *random replay* (resample i.i.d. from a buffer of 2,000 vanilla messages pre-collected from the same checkpoint, preserving the marginal token distribution while destroying conditional dependence on the current state); *permuted* (swap messages across the three predators using a per-timestep random permutation, preserving the multiset of emitted messages but misaligning sender identity with content); and *Gaussian noise* (add zero-mean Gaussian noise with $\sigma = 0.5$ and clip to the one-hot range, applying a matched-marginal perturbation without removing the underlying signal). Full mathematical definitions are given in Appendix A.1.

A causally communicating policy should exhibit return ordering $\text{vanilla} > \text{random replay} \approx \text{zero} > \text{permuted}$, with permuted being the most damaging because it actively misaligns sender and content Lowe et al. [2019]. Departures from this ordering indicate that the listener policy is insensitive to message content beyond coarse presence-versus-absence.

For each (checkpoint, intervention, opponent) cell we evaluate 200 episodes of length 100 with deterministic predator actions, against both the random-prey and heuristic-prey opponents. We report return change $\Delta R = R_{\text{intervention}} - R_{\text{vanilla}}$, with mean and standard deviation across the three training seeds, together with capture rate, first-capture step, and per-episode predator-landmark collision count. Capture rate is included as an upper-bound check: an intervention that preserves capture rate near 1.0 but degrades return reflects efficiency loss, not task failure.

4.2.3 Communication Efficiency (Per-Speak Cost)

To probe whether the trained policy uses the channel out of necessity or by default, we add a small per-speak cost to the reward and sweep its magnitude. The shaped reward is

$$r'_{i,t} = r_{i,t} - \lambda \cdot \mathbb{I}(c_{i,t} \neq 0), \quad (1)$$

where $\mathbb{I}(c_{i,t} \neq 0)$ is 1 when agent i emits a non-null token at time t and 0 otherwise, and $\lambda \geq 0$ is the speak-cost coefficient. This is an action-level reward-shaping term in the sense of Ng et al. [1999]: r' augments r with an action-dependent term while the underlying task signal in r is preserved. The null token (§4.2.1) is required for the cost to provide a meaningful learning signal: without a silent action, the policy cannot reduce the cost by changing behavior.

We sweep λ on a $\sqrt{2}$ -spaced geometric grid,

$$\lambda \in \{0, 0.5, 0.71, 1.0, 1.41, 2.0, 10.0\},$$

chosen to densely sample the transition between $\lambda = 0$ (no cost) and $\lambda \approx 1$ (cost magnitude comparable to the per-step movement reward), with $\lambda = 10$ included as an extreme silence-enforcing condition. For each λ we train three seeds from scratch on comm8. At evaluation time, no cost is applied; all checkpoints are evaluated under the unshaped task reward r so returns at different λ are directly comparable. Implementation details are in Appendix A.2.

4.2.4 Communication Content (Probing Analysis)

To complement the necessity tests in §4.2.2, we apply linear probing to the trained communication tokens to ask not whether the channel improves return, but whether a simple decoder can recover task-relevant variables from the token alone Alain and Bengio [2016], Hewitt and Liang [2019]. We freeze a trained comm8 checkpoint and roll out 500 episodes of length 100 (50,000 predator-timesteps) against the heuristic prey, logging each one-hot token together with simulator-derived ground-truth labels. The state trajectory is always driven by the trained policy; control variants (below) are evaluated on the *same* trajectory so that probe-accuracy differences reflect message content rather than state coverage.

We probe four labels per (predator, timestep): (i) relative prey position $(\Delta x, \Delta y) \in \mathbb{R}^2$, fit by OLS regression and scored by R^2 ; (ii) prey quadrant (4-way), scored by balanced accuracy; (iii)

self-is-closest (binary), scored by AUROC; and (iv) *closest-predator identity* (3-way), a team-level target whose probe input is the concatenation $[m_1, m_2, m_3]$. Classification probes use L_2 -regularized logistic regression with class-balanced weights. Per-wolf probes for (i)–(iii) are fit independently on each predator; reported scores are mean $\pm$ std across the three predators.

For each target we also report two control conditions [Hewitt and Liang, 2019]: **random-init**, messages from same-architecture but randomly-initialized communicators (3 seeds, evaluated on the same state trajectory), isolating the contribution of architecture and observation structure; and **marginal**, messages drawn i.i.d. from each predator’s empirical token-frequency distribution, isolating the contribution of token-frequency imbalance. We split the 500 episodes 70/15/15 into train/val/test at the episode level to prevent within-episode leakage; the val partition is reserved for future probe variants requiring hyperparameter tuning. The probing analysis supports a content-encoding claim when $\text{score}_{\text{trained}} \gg \text{score}_{\text{random-init}} \approx \text{score}_{\text{marginal}}$.

As a non-parametric complement, we also report mutual information $I(m_{i,t}; y)$ between each predator’s token and three discrete labels (prey quadrant, self-is-closest, distance bin), computed as a plug-in estimator from empirical joint counts.

4.2.5 Role Emergence (Behavioral Clustering)

The probing analysis in §4.2.4 asks whether messages encode task-relevant information. We complement it with a behavioral test asking whether predators act out differentiated roles, and whether such structure depends on the communication channel.

We define a pursuit frame at each timestep as the unit vector $\hat{\mathbf{f}}_t$ from the predator centroid toward the prey, and let $\Delta p_t^i = p_t^i - p_t^{\text{prey}}$. For each predator we extract five features: the prey distance and its rank among the three predators, the signed front-back projection $\Delta p_t^i \cdot \hat{\mathbf{f}}_t$, and the signed and absolute lateral offsets perpendicular to $\hat{\mathbf{f}}_t$. Features are standardized using statistics fit on the *vanilla* rollouts. We fit K -means with $K = 3$ (matching the predator team size) on standardized descriptors from 200-episode *vanilla* rollouts under the heuristic prey (120,000 samples per checkpoint), and reuse the same scaler and centroids to project *zero-message* rollouts so cluster identity stays comparable.

Clusters are labeled post-hoc by centroid geometry: the cluster minimizing (distance_rank, distance) is *Chaser*; among the remaining two, the one with larger mean |lateral| is *Flanker*; the last is *Support*.

For every (checkpoint, intervention) cell we compute silhouette, Davies–Bouldin, Calinski–Harabasz, and normalized within-cluster inertia, evaluated under both pre-action and post-action states. The main text reports silhouette (Table 5); other metrics agree in direction and are deferred to the appendix. We compare three checkpoints — *no-comm*, *comm8*, and *comm32* — under *vanilla* and *zero-message* conditions, the latter matching the *zero* intervention in §4.2.2. To quantify whether the channel labels roles, we compute the normalized mutual information $\text{NMI}(c_t^i, r_t^i)$ between each predator’s emitted token (arg max of the communication output, with the all-zero vector mapped to a silent symbol) and its assigned cluster on the same rollouts. We also report per-predator role-occupancy fractions and role self-transition probabilities as descriptive statistics for specialization and temporal stability.

5 Experiments

5.1 Bandwidth Sweep Results

We first ask whether the communication channel contributes to coordination, and how its contribution scales with bandwidth d . Table 2 reports predator return averaged over three independently trained seeds for each $d \in \{0, 2, 4, 8, 16, 32, 64\}$, evaluated against the random and heuristic prey opponents defined in Section 4.2.1. Figure 1 plots the same data with seed-level error bars. Capture rate saturates at ≥ 0.99 across all $d \geq 2$ and is therefore omitted from the main display.

Adding any communication channel produces a substantial return improvement over the $d = 0$ baseline against both opponents (Table 2, Figure 1), well above seed-level noise. The improvement is larger against the weaker random opponent than against the heuristic opponent, but consistent in sign

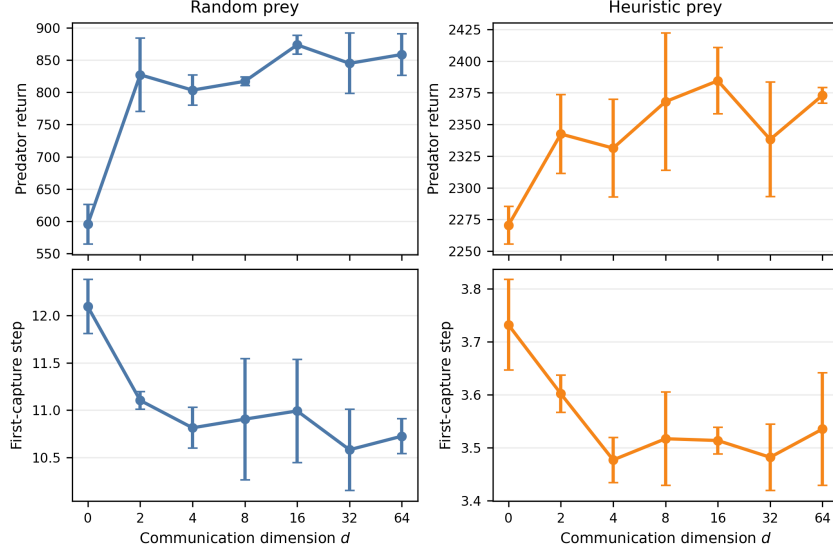


Figure 1: Predator return (top) and first-capture step (bottom) as a function of communication dimension d .

Table 2: Predator return under fixed-opponent evaluation, reported as mean $\pm$ standard deviation across three training seeds. Δ columns report return improvement relative to the no-communication baseline $d = 0$. The best mean return in each opponent setting is shown in bold.

d	Random return	Δ vs. $d = 0$	Heuristic return	Δ vs. $d = 0$
0	595.6 $\pm$ 30.7	–	2270.6 $\pm$ 14.8	–
2	827.2 $\pm$ 56.8	+231.6	2342.5 $\pm$ 31.1	+71.9
4	803.4 $\pm$ 23.4	+207.8	2331.3 $\pm$ 38.4	+60.8
8	817.1 $\pm$ 6.8	+221.5	2368.0 $\pm$ 54.1	+97.4
16	873.6 $\pm$ 14.7	+278.0	2384.5 $\pm$ 26.1	+113.9
32	844.9 $\pm$ 46.9	+249.3	2338.2 $\pm$ 45.2	+67.6
64	858.7 $\pm$ 32.2	+263.1	2373.0 $\pm$ 6.2	+102.4

across all $d \geq 2$. These effects indicate that the channel contributes meaningfully to pursuit efficiency, even though capture rate alone, saturating near 1.0 across all communication-enabled conditions, does not reveal this difference.

Returns do not increase monotonically with d . The highest mean return is observed at $d = 16$ for both opponents, but standard deviations across $d \in \{8, 16, 32, 64\}$ overlap substantially and the gap between $d = 16$ and $d = 2$ is moderate relative to within-condition seed variability. We therefore report $d = 16$ as the best observed configuration while avoiding a claim of statistical optimality. The qualitative pattern suggests that performance gains saturate after approximately $d = 8$, consistent with diminishing returns above moderate bandwidths in continuous-broadcast communication.

5.2 Counterfactual Intervention Battery Results

The bandwidth sweep establishes that adding a communication channel improves return, but it does not determine whether the trained policy actually uses message contents at evaluation time. A movement policy that solves the task from local observations could yield the same training-time gain while the message head remains a passive co-product. To test causal necessity, we apply the four counterfactual interventions from Section 4.2.2 to checkpoints at each $d \geq 2$: zero, which removes information; random replay and permutation, which preserve marginal statistics while destroying conditional structure; and Gaussian noise, which applies a matched marginal perturbation. Table 3 reports the within-checkpoint return change $\Delta R = R_{\text{intervention}} - R_{\text{vanilla}}$, averaged across bandwidths and seeds. Figure 2 displays the same data as a bandwidth-by-intervention heatmap.

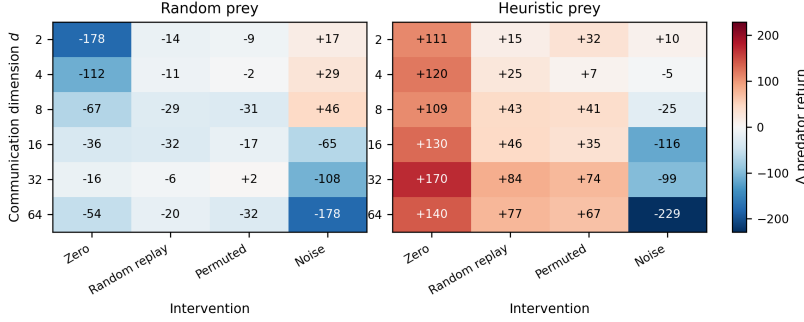


Figure 2: Bandwidth by Intervention Heatmap

Table 3: Within-checkpoint intervention effects relative to vanilla evaluation, averaged over bandwidths $d \geq 2$ and three training seeds. Negative ΔR indicates degraded return under intervention. Capture-step and collision-step changes are reported in environment timesteps.

Opponent	Intervention	Δ return	Δ capture step	Δ collision
Random	Zero	-77.2 ± 60.9	+1.13	-3.58
Random	Random replay	-18.5 ± 20.4	+0.52	-1.15
Random	Permuted	-14.8 ± 21.2	-0.26	-0.90
Random	Noise	-43.3 ± 90.4	+0.42	-1.50
Heuristic	Zero	$+130.0 \pm 37.5$	+0.10	+0.72
Heuristic	Random replay	$+48.2 \pm 34.0$	+0.05	-0.12
Heuristic	Permuted	$+43.0 \pm 33.0$	+0.03	-0.06
Heuristic	Noise	-77.4 ± 95.2	+0.09	-1.31

Capture rate is preserved under every intervention. Across all $d \geq 2$ and both opponents, capture rate remains saturated at ≥ 0.99 under every intervention. Intervention effects therefore appear as efficiency changes in return, first-capture step, and sustained collision behavior, rather than as task failure. The channel is not required for task success on `simple_tag` against either opponent, even though removing it can slow or alter pursuit. This is a strictly weaker claim than full causal necessity.

The random-prey ordering contradicts the Lowe et al. diagnostic. A causally communicating policy should exhibit the ordering vanilla > random replay $\approx$ zero > permuted, with the permuted condition most damaging because it actively misaligns sender identity with message content. We instead observe vanilla > zero $\approx$ noise > random replay $\approx$ permuted (Table 3). The information-removing zero intervention is more damaging than the information-corrupting permuted or random-replay interventions, both of which preserve marginal statistics while destroying conditional structure. This indicates that the listener policy is largely insensitive to message content beyond coarse presence-versus-absence; only the all-zero condition produces a robust degradation, consistent with the policy responding to the absence of a habitually present input rather than to its specific content.

Heuristic-opponent results require a separate interpretation. Against heuristic prey, three of the four interventions *increase* mean return relative to vanilla, and only Gaussian noise produces a decrease (Table 3). This counterintuitive pattern is consistent with `simple_tag` return being sensitive to collision dynamics, where sustained contact contributes positively to return regardless of whether predators are coordinating semantically. On the more evasive opponent, message intervention is therefore not equivalent to disrupting coordination, and we rely primarily on the random-opponent results for the necessity claim. The probing analysis in Section 5.3 revisits this question from the content side.

5.3 Channel Content via Probing

We now ask, independently of the tests in Sections 5.2 and 5.4, what task-relevant information the trained channel encodes. Following the probing protocol in Section 4.2.4, we collect 50,000 message-ground-truth pairs from the `comm8` checkpoint and fit linear probes for four target labels: relative

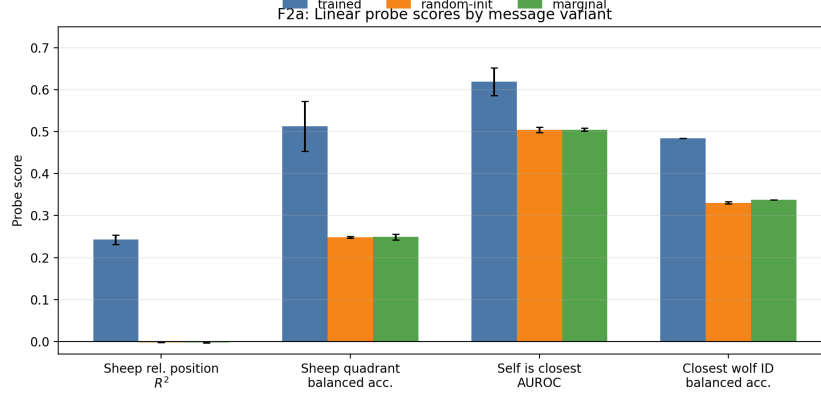


Figure 3: Linear probe scores for four task-relevant labels under three message-source conditions (trained, random-init, marginal). Trained messages substantially exceed both controls on prey-spatial tasks (relative position, quadrant), with smaller margins on role-related tasks (self-is-closest, closest wolf ID). Error bars show std across three predators.

Table 4: Information-theoretic summary of trained communication tokens. Entropy is measured per predator over the 8-token vocabulary; mutual information is reported in bits between each predator’s token and discrete geometric labels.

Predator	Token entropy	MI quadrant	MI self-closest	MI distance-bin
wolf0	2.330 / 3.0	0.364	0.045	0.023
wolf1	1.822 / 3.0	0.397	0.029	0.006
wolf2	2.161 / 3.0	0.524	0.051	0.014

prey position regression, prey quadrant classification, `is_self_closest` binary classification, and `closest_wolf_id` three-way classification. The random-init and marginal message variants serve as selectivity controls, isolating the contributions of architecture and token-frequency bias respectively. Figure 3 reports probe scores; full numerical values appear in Appendix Table 7.

Spatial information is robustly encoded. Trained messages substantially exceed both controls on every spatial-prediction task (Figure 3). Prey-quadrant balanced accuracy reaches roughly twice the chance level of 0.25, and linear probes recover the relative prey position ($\Delta x_i, \Delta y_i$) with strictly positive R^2 , while both controls remain near chance ($R^2 \approx 0$). The channel therefore encodes spatial structure beyond what architecture or token-frequency bias alone can supply.

Role information is weaker. Role-related labels are also recoverable but with smaller margins: `is_self_closest` AUROC modestly exceeds 0.5, and the team-level `closest_wolf_id` probe exceeds its 0.33 chance baseline (Table 7). Mutual information confirms this asymmetry: $I(m^k; \text{quadrant})$ is roughly an order of magnitude larger than $I(m^k; \text{is_self_closest})$ for every predator (Table 4). The channel thus carries weaker signal about *who* should pursue than about *where* the prey is.

Content is redundant with observation. This content-side finding sits in tension with Sections 5.2 and 5.4: the channel encodes prey-spatial information that linear probes can recover, yet zeroing it does not measurably degrade capture rate, role separability, or message-role NMI. The most parsimonious reconciliation is that under near-full observability, each predator already recovers the same spatial information from its own state, so the channel’s content is measurable but not behaviorally necessary.

5.4 Role Emergence Results

We next examine whether the channel drives *behavioral* coordination, distinct from the performance test in Section 5.2 and the content test in Section 5.3. Following the protocol in Section 4.2.5, we collect 200-episode rollouts under heuristic prey for three checkpoints (no-comm, comm8, comm32),

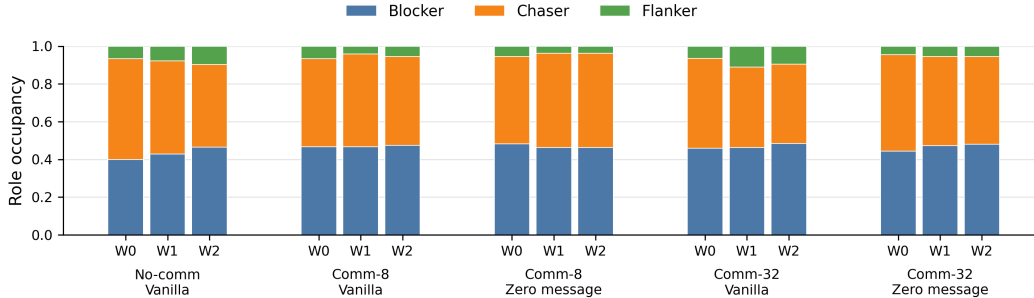


Figure 4: Role occupancy across communication settings under heuristic prey. Each stacked bar shows the fraction of predator timesteps assigned to each geometry-derived role for one wolf. Similar occupancy patterns in no-communication, vanilla communication, and zero-message rollouts indicate that role-like organization is stable across communication bandwidths and does not collapse when messages are removed.

Table 5: Role-cluster separability across communication conditions. Silhouette is higher when clusters are better separated. Message-role NMI is computed on token-role pairs from the same rollouts. *Pre* and *post* denote pre-action and post-action state evaluation, respectively.

Model	State	Vanilla silhouette	Zero silhouette	Message-role NMI
no-comm	post	0.357	–	–
no-comm	pre	0.361	–	–
comm8	post	0.356	0.362	0.0087
comm8	pre	0.358	0.364	0.0102
comm32	post	0.354	0.373	0.0251
comm32	pre	0.355	0.374	0.0272

yielding 120,000 predator-timestep samples per checkpoint. We fit K -means with $K = 3$ on standardized pursuit-geometry descriptors from the vanilla rollouts, and reuse the same cluster basis on zero-message rollouts. Table 5 reports cluster separability and Figure 4 the role-occupancy decomposition.

Role-like geometry emerges, but does not depend on communication. K -means recovers three clusters labeled *Chaser*, *Support*, and *Flanker* by centroid geometry,¹ with consistent semantics across communication settings. Two tests show this structure is not communication-driven (Table 5): the no-comm silhouette scores closely match those of comm8 and comm32, and zeroing messages in trained communicators leaves separability unchanged or slightly improved. Davies–Bouldin moves in the same direction. If communication causally sustained role organization, zeroing the channel should degrade these metrics; we observe the opposite.

Messages do not encode role identity. Message-role normalized mutual information is near zero for both comm8 and comm32 (Table 5); the slightly higher comm32 value is partly inflated by token-frequency sparsity, where rare tokens map deterministically to a single role by sample-size accident. High-frequency tokens, in contrast, do not function as role identifiers: the most common comm32 token (frequency 16.3%) splits across Chaser/Support/Flanker close to the marginal role occupancy.

Roles are dynamic rather than agent-specific. Per-predator role occupancy is similar across the three predators in every condition (Figure 4); no predator preferentially occupies a single role. Combined with role self-transition probabilities in the range 0.80 to 0.84, this indicates that roles are reassigned dynamically based on instantaneous pursuit geometry rather than fixed agent specialization. The predators therefore exhibit role-like behavior without role-like identity.

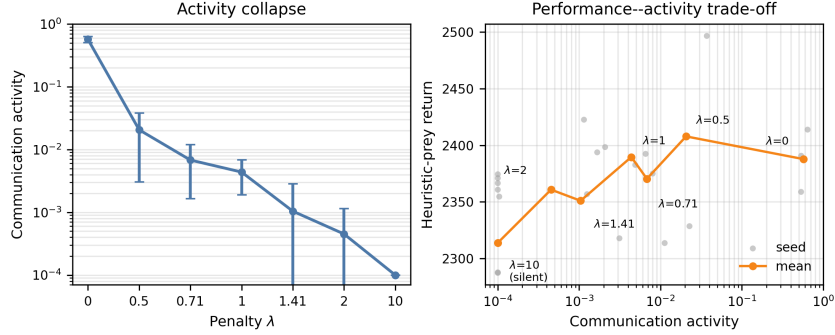


Figure 5: Communication-activity sweep for comm8. Left: mean speaking activity decreases sharply as the activity penalty λ increases, with a phase-like collapse between $\lambda = 0$ and $\lambda = 0.5$. Right: Pareto-style performance-activity trade-off under heuristic prey. Gray points denote individual seeds and the orange curve denotes the seed mean. Small penalties greatly reduce communication activity while preserving fixed-opponent return, indicating that the learned protocol is highly compressible.

Table 6: Communication activity sweep for comm8. Small speaking penalties preserve task performance while sharply reducing nonzero communication activity. Returns and steps are reported as mean $\pm$ standard deviation over three seeds. Activity is the fraction of timestep-agent pairs with nonzero messages.

Penalty λ	Random return	Heuristic return	Heuristic Δ return	Comm. activity	Token-0 share
0	823.8 ± 51.1	2387.8 ± 27.6	0	0.571 ± 0.062	0.429
0.5	829.4 ± 48.5	2407.8 ± 84.4	$+20.0 \pm 104.6$	0.021 ± 0.018	0.979
1.0	833.1 ± 23.3	2389.6 ± 6.0	$+1.8 \pm 22.6$	0.0044 ± 0.0025	0.996
10.0	800.0 ± 43.5	2313.9 ± 45.6	-73.9 ± 28.2	0.0000 ± 0.0000	1.000

5.5 Bandwidth Efficiency via Communication Activity Sweep

Speaking activity collapses while task performance is preserved. Figure 5 reports the Pareto trade-off between heuristic-prey return and communication activity over the seven λ values in the comm8 sweep. The activity rate undergoes a sharp phase transition between $\lambda = 0$ and $\lambda = 0.5$ (a more than 20-fold reduction), and falls below 10^{-4} at $\lambda = 10$. Task return remains within seed variance of the unpenalized baseline across the pre-collapse and immediate post-collapse regimes; statistically significant degradation appears only at $\lambda = 10$ (paired t -test, $p < 0.05$). Mild speaking costs therefore collapse communication without measurable performance loss, while only strongly enforced silence sacrifices return.

Together with the test-time cut-off intervention (Section 5.2) and the no-communication baseline (Section 5.1), the activity sweep gives a stronger training-time result: even a small per-speak cost drives the policy into a near-silent regime while recovering full task performance. The three diagnostics jointly suggest that the channel is used opportunistically but is not required for the task; coordination is mainly supported by the movement policy and positional observations.

Two additional statistics rule out the possibility that the activity sweep merely uncovers an already sparse but structured protocol. In the unpenalized comm8 policy, active messages are broadly distributed rather than concentrated on a single symbol or specialized by agent: conditional active-symbol entropy is close to the $\log_2 8 = 3$ -bit uniform ceiling, no symbol exceeds roughly 28% probability, and several symbols carry non-trivial mass. Under $\lambda = 1.0$, communication is almost entirely suppressed, leaving too few active messages to support a stable protocol. These results are consistent with Lowe et al. [2019]’s positive-signaling diagnostic for non-functional channels: messages remain statistically present without clear task-relevant specialization, and moderate penalty collapses the channel without recovering a compressed code.

¹We use *Support* rather than *Blocker* because the front-back coordinate is uniformly negative across all three clusters, indicating that predators predominantly trail the prey rather than encircle it.

6 Conclusions

Using `simple_tag` as the experimental platform, we systematically investigated credit assignment and the role of explicit communication in cooperative MAPPO. Our contributions are twofold.

First, we identified two implementation bugs in the vanilla `simple_tag` reward that conflate per-predator contributions with team-level outcomes, and patched them with a per-agent distance shaping term and a mixed self-credit / team-credit capture scheme, supplemented by a predator-landmark collision penalty. The corrected reward provides differentiated, role-relevant feedback signals and enables stable MAPPO training.

Second, we evaluated communication along three independent diagnostic axes: a bandwidth sweep, test-time counterfactual interventions, and a per-speak cost sweep. The bandwidth sweep showed that adding any channel improves return over the no-communication baseline but saturates after roughly $d = 8$. Counterfactual interventions showed that capture rate remains saturated at ≥ 0.99 under every intervention, and the random-prey ordering does not match the Lowe et al. (2019) prediction for content-dependent protocols. Linear probes recovered prey-spatial information from the trained tokens, but role-cluster geometry persists at full strength even when messages are zeroed, and message-role NMI is near zero. The per-speak cost sweep produced a third independent line of evidence: a small penalty drives speaking activity into a near-silent regime while preserving full task return.

Together, these results indicate that under the near-full observability of `simple_tag`, the channel encodes some task-relevant information but is used opportunistically rather than necessarily; co-ordination is largely supported by the movement policy and positional observations. This finding underscores the importance of environmental complexity and partial observability when evaluating the value of explicit communication mechanisms.

7 Limitations & Future Work

Several limitations qualify our conclusions and motivate future directions. Most importantly, `simple_tag` is fully observable and relatively low-dimensional. Each predator can directly access the positions of others, so messages are largely redundant with local observations; our communication-redundancy finding may not generalize to partially observable or higher-dimensional environments. Evaluating the same diagnostic battery under partial observability or restricted information access (e.g., masked observations) is a natural next step, and would test when communication transitions from optional to essential.

This effect is compounded by MAPPO’s centralized critic, which has access to global state during training and lets the policy implicitly exploit shared information without relying on explicit communication at execution. Decentralized-critic variants would provide a stronger test of whether communication carries causal weight.

Our communication mechanism is also deliberately simple: fixed-dimensional discrete tokens emitted at every step. This may limit the emergence of more sophisticated protocols, so conclusions about channel content should be read as conditional on this design. Future work can explore continuous or learned message embeddings, attention-based communication, and event-triggered emission, paired with deeper analysis tools (e.g., information-theoretic decomposition or causal-influence measures) to quantify how much of a message actually shapes downstream decisions.

Finally, our reward shaping and scaling are manually designed. While this improved training stability and made credit assignment tractable, it introduces inductive bias that may shape the learned behaviors. Learning-based reward design could reduce this bias and improve generalization across tasks.

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A Technical appendices and supplementary material

A.1 Counterfactual Intervention Definitions

Let c_t^i denote the communication slot emitted by the policy for predator i at time t , and $\tilde{c}_t^i$ the post-intervention slot passed to the environment. The five interventions in §4.2.2 are:

Vanilla. $\tilde{c}_t^i = c_t^i$. No intervention; serves as the reference for ΔR .

Table 7: Full linear-probe scores corresponding to Figure 3. Values are mean $\pm$ std.

Target	Trained	Random-init	Marginal
Sheep rel. position (R^2)	0.243 ± 0.011	-0.002 ± 0.000	-0.002 ± 0.002
Sheep quadrant (bal. acc.)	0.513 ± 0.059	0.248 ± 0.002	0.249 ± 0.007
Self is closest (AUROC)	0.619 ± 0.033	0.504 ± 0.006	0.505 ± 0.003
Closest wolf ID (bal. acc.)	0.484 ± 0.000	0.330 ± 0.003	0.338 ± 0.000

Zero. $\tilde{c}_t^i = \mathbf{0}$. The all-zero vector replaces the one-hot token. Note that for bandwidth-sweep checkpoints (no null token), the all-zero vector is itself out-of-distribution: every trained message has exactly one active component, so listeners receive an unfamiliar absence-of-presence signal.

Random replay. $\tilde{c}_t^i \sim \text{Uniform}(\mathcal{B})$, where $\mathcal{B}$ is a buffer of 2,000 vanilla messages pre-collected from the same checkpoint on independent rollouts (i.i.d. uniform sampling, independent across agents and timesteps). Replay preserves the marginal token distribution while destroying the conditional dependence on the current state.

Permuted. $\tilde{c}_t^i = c_t^{\pi(i)}$ for a uniform random permutation π over the three predators, redrawn at every timestep. This preserves the multiset of messages emitted at time t but misaligns sender identity with content.

Gaussian noise. $\tilde{c}_t^i = \text{clip}(c_t^i + \varepsilon, 0, 1)$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$, $\sigma = 0.5$. This adds matched-marginal noise without removing the underlying signal; the clip enforces the one-hot range.

A.2 Per-Speak Cost: Implementation Details

Application scope. The cost in Eq. (1) is applied only to predator (adversary) agents that have a communication channel. The prey reward, the prey policy, and the predator movement-action distribution are unchanged. At evaluation time, no cost is applied: all checkpoints are evaluated under the unshaped task reward r so that returns at different λ are directly comparable.

Implementation. At each environment step we precompute the binary speaking indicator $a_{i,t}^{\text{comm}} = \mathbb{I}(c_{i,t} \neq \mathbf{0})$, multiply by λ , and subtract from the per-agent reward returned by the environment before passing the trajectory to the learner. The shaped reward r' feeds into MAPPO’s advantage and value-function targets; gradient updates for both the movement and communication action heads use r' .

A.3 Role Cluster Geometry (Vanilla Rollouts)

Figure 6 visualizes the K -means clustering described in §4.2.5 by projecting the five-dimensional pursuit-geometry descriptors onto the two most discriminative axes: the signed front-back projection $\Delta p_t^i \cdot \hat{f}_t$ (horizontal) and the signed lateral offset perpendicular to $\hat{f}_t$ (vertical). Each point is one (predator, timestep) pair from 200-episode vanilla rollouts on the `comm8` checkpoint under the heuristic prey (120,000 samples total), colored by the cluster assignment used throughout §5.4.

The three clusters separate cleanly in this 2-D projection and the visual geometry agrees with the centroid-based labels. The Chaser cluster (orange) concentrates near `front_back` ≈ 0 with small `|lateral|`, corresponding to the predator that has closed the gap to the prey along the pursuit axis. The cluster relabeled *Support* in the main text (blue; the figure legend retains the script’s default name “Blocker”) sits at moderately negative front-back values with limited lateral spread, indicating predators that trail the prey along the pursuit direction without flanking. The Flanker cluster (green) extends to substantially more negative front-back with a much larger lateral dispersion, consistent with predators positioned off to the side of the centroid–prey axis. Crucially, the bulk of every cluster — and in particular all three cluster centroids — sits at `front_back` ≤ 0 , motivating the relabeling of Blocker to Support: across all conditions the predators predominantly trail rather than encircle the prey, so the near-axis trailing cluster is better described as a supporting role behind the chaser than as an encircling blocker.

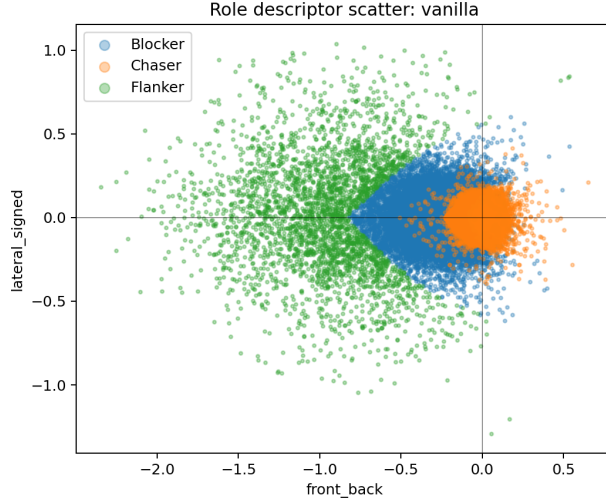


Figure 6: Role descriptor scatter in the (front-back, lateral-signed) plane on vanilla comm8 rollouts under heuristic prey. Colors denote the K -means cluster assignment ($K = 3$)



Figure 7: Training visualization of agent behaviors in the simple_tag environment under communication dimension $K = 2$.

B Supplementary material

B.1 Result rendering example

From the training visualization, we can observe that the agents exhibit clear cooperative behaviors with implicit role differentiation. In particular, the alternating black and white circles among predators indicate active communication, suggesting that agents are exchanging information to coordinate their movements during pursuit.